MILITARY SPECIFICATION

DETECTORS, PIN, FIBER OPTIC,

GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for positive-intrinsic-negative (PIN) fiber optic detectors. These optical detectors contain optical windows or integrally attached pigtail fibers for coupling optical power from the fiber optic link fiber into the detector. Pigtail fibers specified herein are mechanically and optically compatible with the fibers specified in DOD-C-85045 and also with the connectors specified in MIL-C-28876.

1.2 Classification. PIN detectors are classified into two product assurance quality levels, A and B (QLA and QLB). QLA parts are hermetic devices. QLB parts are nonhermetic devices.

1.2.1 Military part number. The military part number shall consist of the letter "D", the basic number of the associated detail specification, and an assigned dash number as shown in tables I, II, and III:

D	24620/01	A	08	A
Military designator	Detail specification number	Sealing level designator (table I)	Wavelength designator (table II)	Pigtail fiber designator (table III)

TABLE I. Sealing level designator

Symbol	Sealing level
A	Hermetic
B	Nonhermetic

TABLE II. Wavelength designator.

Symbol	Wavelength of operation in nanometers
08	820 - 910
13	1250 - 1350
15	1400 - 1600

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command (SEA 55Z3), DOD Standardization Program and Documents Division, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE III. Pigtail fiber designator.

Symbol	Core/cladding diameter (µm)	Core index profile
A	50/125	step
B	50/125	graded
C	100/140	step
D	100/140	graded

Note - If the detector is not a pigtailed device, then the pigtail fiber designator symbol is deleted from the part number.

For example, part number D 24620/1 - A13B identifies a glass-pigtailed type, pin, fiber optic detector that is hermetic, operating at a wavelength of 1250-1350 nanometers, and using a graded index core fiber with core/cladding diameters of 50/125 micrometers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standard (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-19491 - Semiconductor Devices, Packaging of

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-750 - Test Methods for Semiconductor Devices
MIL-STD-889 - Dissimilar Metals
MIL-STD-1285 - Marking of Electrical and Electronic Parts
DOD-STD-1678 - Fiber Optic Test Methods and Instrumentation.
DOD-STD-1686 - Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric).
MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be as specified herein and in accordance with the applicable associated detail specifications. In the event of any conflict between requirements of this specification and the associated detail specifications, the latter shall govern.

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3.2 Qualification. Detectors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the optic detectors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product. Contractors shall submit a certificate of compliance with the specified material requirements (see 6.2.2).

3.3.1 Metals. External metal surfaces shall be of corrosion resistant materials or shall be plated or treated to resist corrosion in accordance with MIL-STD-889.

3.3.2 Pigtails. Pigtails shall be optical glass fibers as specified (see 3.1).

3.3.3 Recovered materials. Unless otherwise specified herein, material shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specificaton unless otherwise specifically specified.

3.4 Design and construction. Four detector design configurations are specified as follows.

a. A detector with sealed optical window where input coupling is obtained by butt-coupling the link fiber to the window. This is a hermetic (QLA) detector arrangement.
b. A detector with sealed optical window where input coupling is obtained through a pigtail fiber cemented to the window. This is a hermetic (QLA) detector arrangement.
c. A detector with a sealed fiber pigtail integrally attached to the detector chip active region by "graded-seal" methods. This is a hermetic (QLA) detector arrangement.
d. A detector with a fiber pigtail epoxied or otherwise cemented to the detector chip active region. This is a nonhermetic (QLB) detector arrangement.

3.4.1 Visual and mechanical inspection. When tested in accordance with 4.6.1.1, the detector shall meet the lot tolerance percent defective (LTPD) requirements specified in tables IV, V and VII.

3.4.2 Physical dimensions. When tested in accordance with 4.6.1.2, the detector shall meet the LTPD requirement specified in table V and VII. Overall dimensions, dimensional tolerances and physical characteristics for these detectors shall be as specified (see 3.1). Pigtailed devices shall use a single pigtail fiber with one end integrally attached to the detector. The other pigtail end shall be terminated or unterminated as specified (see 3.1). The pigtail fiber shall be sheathed in materials and of constructions as specified (see 3.1). The detector shall accept input electrical signals which provide bias for proper detector operation, and shall receive optical power through its window or attached pigtail fiber as specified (see 3.1). The numerical aperture of the pigtail fiber and detector mounting characteristics shall be as specified (see 3.1). Detector lead-in conductors shall be composed of materials and of dimensions adequate for their specified maximum current carrying capacity.

3.4.3 General, optical, electrical and electro-optical. Requirements specified herein and in the associated detail specification (see 3.1) shall be used to determine acceptance or rejection for detectors subjected to the test method exposures of 4.6.3. End-point parameters (see 6.4.3) are listed with requirement and test paragraphs specified in table IV (see 4.5.6).

3.4.3.1 Operating and storage temperature ranges. The maximum operating and storage temperature range shall be -55 C to +105 C.

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3.4.3.2 Forward current. The maximum forward current under pulse and continuous operating conditions shall be as specified (see 3.1).

3.4.3 3 Power dissipation. The maximum power dissipation shall be as specified (see 3.1).

3.4.3.4 Reverse voltage. The maximum reverse voltage (bias) shall be as specified (see 3.1).

3.4.3.5 Optical end points. Optical end-point levels, LTPD's and delta's (see 6.4) shall be as specified (see 3.1).

3.4.3.6 Electrical end points. Electrical end-point levels, LTPD's and delta's (see 6.4) shall be as specified (see 3.1).

3.4.4 Detector input port protection. Detector optical input ports shall be capped (covered) at all times other than when under test.

3.5 Performance.

3.5.1 Optical/electrical.

3.5.1.1 Spectral responsivity and quantum efficiency. When tested in accordance with 4.6.2.1, spectral responsivity and quantum efficiency shall be not less than the level specified (see 3.1).

3.5.1.2 Capacitance. When tested in accordance with 4.6.2.2, PIN junction capacitance shall be as specified (see 3.1).

3.5.1.3 Reverse current leakage and breakdown voltage. When tested in accordance with 4.6.2.3, the reverse current leakage level shall not exceed the value specified (see 3.1), and the reverse breakdown voltage shall exceed the voltage value specified in the specification sheet for the specified reverse current level.

3.5.1.4 Dark current. When tested in accordance with 4.6.2.4, the measured dark current shall not exceed the value specified (see 3.1).

3.5.1.5 Responsivity linearity. When tested in accordance with 4.6.2.5, the responsivity linearity shall meet the level specified (see 3.1).

3.5.1.6 Pulse response. When tested in accordance with 4.6.2.6, the pulse rise time (between 10 percent and 90 percent maximum pulse power amplitude) and fall time (between 90 percent and 10 percent maximum pulse power amplitude) shall not exceed the value specified (see 3.1).

3.5.1.7 Analog bandwidth. When tested in accordance with 4.6.2.7, the analog bandwidth shall meet the level specified (see 3.1).

3.5.1.8 Noise equivalent power. When tested in accordance with 4.6.2.8, the noise equivalent power (NEP) shall not exceed the level specified (see 3.1).

3.5.2 Environmental.

3.5.2.1 High temperature (nonoperating) life (LTPD). When tested in accordance with 4.6.3.1 and the visual inspection of 4.6.1.1, the LTPD requirement of table VI, and the end-point parameters and parameter limits specified shall be met (see 3.1).

3.5.2.2 Thermal shock (temperature cycling). When tested in accordance with 4.6.3.2, there shall be no visible evidence of detector marking impairment or other damage. The LTPD requirement of table VI, and the end-point parameters and parameter limits specified shall be met (see 3.1).

3.5.2.3 Constant acceleration. When tested in accordance with 4.6.3.3, the detector shall meet the LTPD requirement of table VII, and the end-point parameters and parmeter limits specified (see 3.1).

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3.5.2.4 Burn-in. When tested in accordance with 4.6.3.4, the detector shall meet the reverse current requirement, and optical parameters and limits specified (see 3.1).

3.5.2.5 Hermetic seal. When tested in accordance with 4.6.3.5, the detector shall meet the LTPD requirements of tables VI and VII and shall not exceed the leakage rates specified (see 3.1).

3.5.2.6 Solderability. When tested in accordance with 4.6.3.6, the detector's electrical leads shall meet the solderability criteria stated in this test method, and meet the LTPD requirement specified in table VI.

3.5.2.7 Resistance to solvents. When tested in accordance with 4.6.3.7, the detector's identification marking shall remain legible and intact. The LTPD requirement specified in table VI shall be met.

3.5.2.8 Decap internal visual design verification. When tested in accordance with 4.6.3.8, the detector shall meet the LTPD requirement specified in table VI.

3.5.2.9 Bond strength. When tested in accordance with 4.6.3.9, the detector shall meet the LTPD requirement specified in table VI.

3.5.2.10 Thermal shock (glass strain). When tested in accordance with 4.6.3.10, the detector shall meet the LTPD requirement specified in table VII.

3.5.2.11 Terminal strength. When tested in accordance with 4.6.3.11, the terminal strength tests for tension and torque shall be applied to both the external electrical leads and the optical pigtail of the detector. The tensile and torque test loads to be applied shall be as specified (see 3.1). Evidence of breakage, loosening or relative motion between the terminal lead or pigtail and the detector body shall constitute failure and part rejection. The LTPD requirement of table VII shall be met.

3.5.2.11.1 Tension. The terminal tensile strength shall be tested in accordance with 4.6.3.11.1. The minimum acceptable tensile load and the duration of application of this load shall be as specified (see 3.1).

3.5.2.11.2 Torque. The terminal torque stength shall be tested in accordance with 4.6.3.11.2. The minimum acceptable torque load and the duration of application of this load shall be as specified (see 3.1).

3.5.2.12 Moisture resistance. When tested in accordance with 4.6.3.12, the detector shall meet the LTPD requirement of table VII, and the end-point parameter and parameter limits specified (see 3.1).

3.5.2.13 Shock. When tested in accordance with 4.6.3.13, the detector shall meet the LTPD requirement of table VII, and the end-point parameters and parameter limits specified (see 3.1).

3.5.2.14 Vibration, variable frequency. When tested in accordance with 4.6.3.14, the detector shall meet the LTPD requirements specified in table VII, and the end-point parameters and parameter limits specified (see 3.1).

3.5.2.15 Salt atmosphere (corrosion). When tested in accordance with 4.6.3.15, the detector shall not exhibit illegible markings, flaking or pitting of the finish or corrosion that would interfere with the use of the device. The LTPD requirement specified in table VII shall be met.

3.6 Marking. Marking shall be in accordance with MIL-STD-1285. The following legible marking (see 3.1) shall be placed on each optical detector at the time of shipment.

a. Polarity identification.
b. Part number.
c. Manufacturer's designating symbol.
d. Electro-static discharge sensitive symbol, orange lettering or other standard technique per MIL-STD-1686.

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3.6.1 Marking legibility. Marking shall remain legible after all tests. Damage to marking on deliverable detector test samples caused by mechanical fixturing in groups B and C nondestructive tests shall not be cause for lot rejection, but devices with damaged marking shall be remarked to insure legibility prior to shipment.

3.7 Color. Detector external package color shall be as specified (see 3.1).

3.8 Mass. Detector mass shall be as specified (see 3.1).

3.9 Workmanship. Optical detectors shall be manufactured and processed in accordance with good design and sound engineering practice and to the requirements of this specification. The external surface of the detector package, header or flange shall be finished and not have any depression or cavity, unless it is part of the original design. Burrs and sharp edges on the external package and leads shall be removed. External parts, elements or coatings shall not be blistered, cracked, softened or exhibit defects that adversely affect the storage, operation or environmental capabilities of these optical detectors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be estabished and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662. Fiber optic test methods and instrumentation, when specified, shall be in accordance with DOD-STD-1678.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be in accordance with MIL-STD-750.

4.3.1 Formation of inspection lots. Detectors to be qualified shall be assembled into an identifiable inspection lot or collection of inspection sublots.

4.3.1.1 Inspection lot. The total number of detectors that the manufacturer submits at any one time for qualification or quality conformance inspection shall constitute an inspection lot. The inspection lot is submitted to determine compliance with the requirements of the detail specification. Each inspection lot shall consist of detectors of a single type or may consist of a collection of sublots of structurally identical detectors (see 4.3.1.1.2) contained on a single detail specification manufactured on the same production line within a single plant through final seal by the same production technique and to the same detector design with the same material requirements within the same 6-week period. Lot identification shall be maintained from the time the lot is assembled to the time it is accepted or rejected.

4.3.1.1.1 Inspection sublot. An inspection sublot shall consist of a single detector type contained on a single detail specification manufactured on the same production line through final seal by the same fabrication technique and to the same detector design with the same material requirements and within the same 6-week period. Sealing level A and B optical sources are not structurally identical and therefore are independent sublots.

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4.3.1.1.2 Structurally identical detector types. Structurally identical detectors are detectors manufactured on the same production line through final seal by the same fabrication technique and to the same detector design with the same material requirements, and differ only electrically. An example of such structurally identical detectors are optical diodes grouped into different current ratings.

4.3.2 Preservation of lot identity. During all screening, inspection and marking operations, each lot and sublot shall be kept segregated, secure and traceable.

4.3.2.1 Security of completed devices. Marked detectors which have passed all screening and quality conformance requirements shall be retained in a secure area prior to shipment or delivery. Detector inventory shall be controlled by type, quantity, product assurance level, and transaction date. Provision shall be made for surveillance by Government representatives. This requirement applies to the manufacturer and contractors.

4.3.3 Sampling. Statistical sampling for qualification and quality conformance inspection shall be in accordance with the appendix of this specification.

4.3.4 Performance verification tests. Optical and electrical performance verification tests (see 4.7.2) shall be accomplished in a nondestructive manner.

4.3.5 Resubmitted lots. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots. When any lot submitted for qualification or quality conformance inspecion fails any subgroup requirement of groups A, B, or C tests, it may be resubmitted once for that particular subgroup using tightened inspection criteria (as defined in the appendix of this specification). A second resubmission using tightened inspection criteria is permitted only if failure analysis is performed to determine the mechanism of failure for each failed detector from the prior submissions and it is determined that failure is due to:

a. A defect that can be effectively removed by rescreening the entire lot, and that rescreen has been performed, or

b. Random type defects which do not reflect poor basic detector design or poor basic processing procedures. In all instances where analysis of the failed detectors indicates that the failure mechanism is due to poor basic processing procedures, a basic design fault, or nonscreenable defects, the lot shall not be resubmitted.

4.3.6 Method of test. Conditions and methods of test shall be in accordance with MIL-STD-750. The general requirements of MIL-STD-750 apply as specified.

4.3.6.1 Alternative test methods. Other test methods or circuits may be substituted for those specified in MIL-STD-750 provided it is demonstrated to the Government that such a substitution in no way relaxes the requirements of this specification. The schematic wiring diagram of the test equipment shall be made available for checking by the Government. Control and calibration of the test equipment shall be established and documented in accordance with MIL-STD-45662.

4.3.6.2 Procedure in case of test equipment failure or operator error. If a detector is believed to have failed as a result of faulty test equipment or operator error, the failure shall be entered in the test record which shall be submitted to the cognizant Government quality assurance representatives along with a complete explanation verifying why the failure is believed to be invalid. The Government quality assurance representative will then decide whether or not the failure is due to a valid part defect. If the Government quality assurance representative rules that the failure is invalid, a replacement detector from the same inspection lot may be added to the sample. The replacement detector shall be subjected to all those tests to which the discarded detector was subjected prior to its failure and to any remaining specified tests to which the discarded detector was not subjected to prior to its failure.

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4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA) (see 6.3) on sample units produced with equipment and procedures normally used in production. For qualification inspection, sources shall be subjected to screening tests and inspections specified in groups A, B, and C for product assurance levels A or B. All samples subjected to groups B and C must be chosen from a lot which has passed the requirements of group A. The following conditions apply:

a. The required LTPD sample from a sublot of each detector type submitted, shall be tested for each group A subgroup.

b. A sample from one sublot shall be tested for each group B subgroup. A sample detector from each sublot (each device type) shall be submitted to the physical property inspections.

c. A sample from one sublot shall be tested for each group C subgroup.

4.4.1 Inspection routine. For qualification, detectors shall be subjected to screening tests (see 4.4.8) and inspections specified in groups A, B, and C for product assurance levels QLA or QLB. All samples subjected to groups B and C must be chosen from a lot which has passed the requirements of group A. The following conditions apply:

a. The required LTPD sample from a sublot of each detector type submitted shall be tested for each group A subgroup.

b. A sample from one sublot shall be tested for each group B subgroup. A sample detector from each sublot (each device type) shall be submitted to the design verification inspection (see 4.7.1).

c. A sample from one sublot shall be tested for each group C subgroup. At the option of the manufacturer, devices from table VI, group B, subgroup 3, may be continued on in group C, subgroup 6, to achieve 1,000 hours total, or separate samples may be used.

4.4.2 End points. End-point electrical and optical measurements (see 4.7.2) shall be taken and recorded before starting and after completion of all specified tests in the subgroups of groups B and C. Pre-test end-point failures shall be replaced by acceptable devices.

4.4.3 Variables data. Group A variables data and the pre-test and post-test groups B and C end-point variables data shall be included in the test report.

4.4.4 Data submitted. Data for all tests shall be recorded in sufficient detail to verify the test procedures and conditions applied.

4.4.5 Lot size. The qualification inspection lot shall be chosen by the manufacturer and the lot and each sublot shall contain at least twice the number of devices required for qualification.

4.4.5.1 Selection of samples. All samples shall be randomly selected from the qualification inspection lot. After a test has started, the manufacturer may add an additional quantity to the initial sample, but this addition may be done only once for any subgroup and the added samples shall be subjected to all the tests within that subgroup. The total samples (initial and added samples) shall determine the new acceptance number. The total defectives of the initial and second sample shall be additive and shall comply with the specified LTPD or lambda. The manufacturer shall retain sufficient devices from the qualification lot to provide for additional samples.

4.4.5.2 Identification of samples. The authorized Government Quality Assurance Representative may, at his option, mark or authorize the marking of each sample to be subjected to qualification testing in order to distinguish these devices from those not intended for qualification inspection.

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4.4.6 Detector screening. The screening procedure for detectors shall be in accordance with table IV and the applicable associated detail specification. All optical detectors (100 percent) shall be subjected to all applicable screening tests (as specified in table IV) in the sequence shown and shall meet the applicable percent defective allowed (PDA) for the type of optical detector and product assurance level (A or B) specified. Detectors which fail any test criteria in the screening sequence, shall be removed from the lot at the time of the observation or immediately at the conclusion of the test in which the failure was observed.

4.4.6.1 Acceptance criteria. Selected electrical and optical parameters shall be designated in the detail specification as end-point measurements. These parameters may also be compared to determine whether the change during reverse bias burn-in (Δ, delta) is indicative of a lot stability problem. The PDA for each inspection lot submitted to burn-in shall be ten percent on all failures. The manufacturer shall not conduct burn-in in addition to that specified. Delta limits shall be defined in the detail specification. When the PDA applies to delta limits, the delta parameter values measured after burn-in (100 percent screening test) shall be compared with the delta parameter values measured prior to that burn-in.

TABLE IV. Qualification inspection.

100% screen inspections	Visual/mechanical inspection	End-point parameters	Paragraph		Exposure conditions
			Requirement	Test method	
Visual and mechanical inspection	V/M	-	3.4.1	4.6.1.1	Ambient observation
...gh temperature (nonoperating) life (LTPD)	V	-	3.5.2.1	4.6.3.1	24 Hours 125°C-QLA 15°C-QLB
Thermal shock (temperature cycling)	V	-	3.5.2.2	4.6.3.2	-40°C-QLA/QLB 125°C-QLA 85°C-QLB
Constant acceleration	V/M	-	3.5.2.3	4.6.3.3	6 Directions 20,000 G
Burn-in *	-	At λ min. I_R at V_R R NEP	3.5.2.4	4.6.3.4	80% of $V_{R(max)}$ 48 Hours 100°C-QLA 70°C-QLB
Hermetic seal Fine leakage rate	M	Leak > 10^{-8} atm $cm^3/2$	3.5.2.5	4.6.3.5	QLA only
Gross leakage rate	V	Leak > 10^{-3} atm $cm^3/2$			QLB only 85°C

* Electrical and optical measurements shall be taken before and after this test to determine the delta parameters as specified (see 3.1).

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4.4.7 Lot release. The lot from which the qualification samples are selected may be offered for delivery under contract after qualification approval has been granted provided screening and quality conformance requirements are satisifed.

4.4.8 Retention of qualification. To retain qualification, the manufacturer shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of a summary of the results of tests performed for quality conformance inspection (groups A, B, and C), including the number and mode of any subgroup failures. The summary shall include results of all quality conformance inspection tests performed on completed lots during the 12-month period. If the summary of the test results indicate nonconformance with specification requirements, (see 4.5.1 and 4.5.2) and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. The results of tests of all reworked lots shall be identified and accounted for. Failure to submit the report within 50 days after the end of each 12-month period may result in loss of qualification for the product.

4.4.8.1 Nonproduction. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during three consecutive reporting periods there has been no production of a device type or structurally identical types, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product to appropriate testing in accordance with the qualification inspection requirements.

4.5 Quality conformance inspection. Quality conformance inspection shall be conducted in accordance with the requirements of groups A, B, and C for the specified product assurance level. Lot sampling shall be in accordance with the appendix of this specification. If a lot is withdrawn in a state of failing to meet quality conformance requirements and is not resubmitted, it shall be considered a failed lot and reported as such. Each lot shall be subjected to groups A and B inspection. Successful completion of group C quality conformance for a given product assurance level shall satisfy the group C requirements for the tested level and for detectors represented by a structurally identical group. The grouping of structurally identical devices shall be as agreed between the manufacturer and the qualifying activity. Each lot shall be subjected to Groups A and B for delivery. Completion of testing of Group C shall not delay shipment.

4.5.1 Corrective action. If six of ten consecutive lots or if three successive lots of a detector type or structurally identical types are rejected for the same reason, corrective action acceptable to the qualifying activity shall be initiated. Failure to initiate corrective action shall result in removal of products from the qualified products list.

4.5.2 Nonconformance. Lots which fail subgroup requirements of group A, B, or C may be resubmitted for inspection one time only. However, if the lot is not resubmitted or fails resubmission, the lot shall not be shipped. Samples from subsequent lots of the detector types in the structurally identical device grouping represented by a failed group C inspection in the case of group C failures, shall then be subjected to all the tests in the subgroup in which the failure occurred, on a lot-by-lot basis until three successive lots pass the failed subgroup. The testing may then return to periodic testing. A detector type which fails a group C inspection shall not be accepted until the type which failed successfully completes the failed group C subgroups. Other types from the same qualified group represented by the failed type may be accepted provided group C inspection requirements have been satisfied for those detector types.

4.5.3 Group A inspection. Group A inspection shall be performed on each inspection lot and shall consist of visual and mechanical inspection, and electrical and optical tests as specified in table V, and the detail specification. Group A inspections may be performed in any order. If an inspection lot is made up of a collection of sublots, each sublot shall pass group A inspection as specified.

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4.5.4 Group B inspection. Group B inspection shall be performed on each lot. Group B shall be in accordance with table VI and the detail specification for the specified product assurance level. Testing of one detector type sublot in any subgroup shall be considered as complying with the requirements for that subgroup for all types in the lot. Different detector types may be used for each subgroup. This inspection shall be applied only to completed and fully marked detectors from lots which have been subjected to and passed the group A requirements. All tests within a subgroup shall be performed in the order specified.

TABLE V. Group A inspection.

Inspection subgroups	Paragraph		Limit
	Requirement	Test method	LTPD
Subgroup 1			
Pre-exposure conformance inspections			
Physical dimensions	3.4.2	4.6.1.2	5
Mass	3.8	4.6.1.1	
Color	3.7	4.6.1.1	
ID marking	3.6	4.6.1.1	
Workmanship	3.9	4.6.1.1	
Subgroup 2			
DC (static) electrical and optical test at +25°C			
Capacitance	3.5.1.2	4.6.2.2	5
Reverse current leakage and breakdown voltage	3.5.1.3	4.6.2.3	
Dark current	3.5.1.4	4.6.2.4	
Subgroup 3			
DC (static) electrical and optical tests at maximum and minimum rated operating temperatures.			
Reverse current leakage and breakdown voltage	3.5.1.3	4.6.2.3	5
Dark current	3.5.1.4	4.6.2.4	
Subgroup 4			
Dynamic electrical and optical tests at +25°C			
Responsivity linearity	3.5.1.5	4.6.2.5	5
Pulse response	3.5.1.6	4.6.2.6	
Analog bandwidth	3.5.1.7	4.6.2.7	
Noise equivalent power	3.5.1.8	4.6.2.8	

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TABLE VI. Group B inspection.

Inspection subgroups	Conditions of test	Paragraph		Limit
		Requirement	Test method	LTPD
Subgroup 1 1/				
Solderability 3/	MIL-STD-202, method 208	3.5.2.6	4.6.3.6	15
Resistance to solvents	MIL-STD-750, method 1022	3.5.2.7	4.6.3.7	
Subgroup 2 2/				
Thermal shock (temperature cycling)	-40°C	3.5.2.2	4.6.3.2	10
Hermetic seal	+125°C-QLA +85°C-QLB	3.5.2.5	4.6.3.5	
a. Fine leak	QLA only			
b. Gross leak	QLB only			
Subgroup 3				
High temperature (nonoperating) life (LTPD)	340 Hours +125°C-QLA +85°C-QLB	3.5.2.1	4.6.3.1	10
Subgroup 4 2/ 4/				
Steady-state operation Life (LTPD) Reverse Bias (burn-in)	MIL-STD-750, method 1038 Test condition A 80% $V_{R(max)}$ +100°C-QLA +70°C-QLB 48 Hours	3.5.2.4	4.6.3.4	5
Subgroup 5 1/				
Decap internal visual-design verification	MIL-STD-750, method 2075	3.5.2.8	4.6.3.8	15
Bond strength	MIL-STD-750, method 2037	3.5.2.9	4.6.3.9	10

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ Electrical or optical end-point measurements shall be taken after these tests as specified (see 3.1).

3/ The LTPD for solderability tests applies to the number of leads inspected except in no case shall less than three devices be used to provide the number of leads required.

4/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 48 hour life tests may be continued on test to 1000 hours in order to satisfy the group C life test requirements. In such cases, either the 48 hour end point measurements must be made as a basis for group B lot acceptance or the 1000 hours end point measurements shall be used as the basis for both group B and group C acceptance.

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TABLE VII. Group C inspection.

Inspection subgroups	Conditions of test	Paragraph		Limit
		Requirement	Test method	LTPD
Subgroup 1 1/				
Physical dimensions	MIL-STD-750, method 2066	3.4.2	4.6.1.2	15
Subgroup 2 2/				
Thermal shock (glass strain)	MIL-STD-750, method 1056, T_{FLUID} = 85°C QLB	3.5.2.10	4.6.3.10	
Terminal strength	MIL-STD-750, method 2030	3.5.2.11	4.6.3.11	
Hermetic seal a. Fine leak b. Gross leak	MIL-STD-750, method 1071 Condition H-QLA Condition C-QLB	3.5.2.5	4.6.3.5	10
Moisture resistance	MIL-STD-750, method 1021	3.5.2.12	4.6.3.12	
Visual and mechanical inspection	MIL-STD-750, method 2071	3.4.1	4.6.1.1	
Subgroup 3 2/				
Shock	MIL-STD-750, method 2016	3.5.2.13	4.6.3.13	10
Vibration (var. frequency)	MIL-STD-750, method 2056	3.5.2.14	4.6.3.14	
Constant acceleration	MIL-STD-750, method 2006	3.5.2.3	4.6.3.3	
Subgroup 4 1/				
Salt atmosphere (corrosion)	MIL-STD-750, method 1041	3.5.2.15	4.6.3.15	15
Subgroup 5 2/ 3/				
Burn-in	MIL-STD-750, method 1038 Test condition A 1000 hours	3.5.2.4	4.6.3.4	5

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ Electrical or optical end-point measurements shall be taken after these tests as specified (see 3.1).

3/ If a given inspection lot undergoing group B inspection has been selected to satisfy group C inspection requirements, the 48 hour life tests may be continued on test to 1,000 hours in order to satisfy the group C life test requirements. In such cases, either the 48 hour end point measurements must be made as a basis for group B lot acceptance or the 1,000 hours end-point measurements shall be used as the basis for both group B and group C acceptance.

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4.5.5 Group C inspection. Group C inspection shall be in accordance with table VII and shall include those tests specified which are performed periodically at six month intervals on at least one detector type from each structurally identical device grouping (from the same or different detail specification) in which the manufacturer has qualified detector types. This inspection shall be applied only to completed and fully marked samples that have satisfied the specified group A LTPD requirements. Samples chosen for Group C shall have passed both Groups A and B.

4.5.5.1 Group C sample selection. Samples for subgroups in group C shall be chosen at random from the first lot submitted for quality conformance inspections during the specified group C inspection interval. Testing of one detector type for each subgroup shall be considered as complying with the requirements for that subgroup for all types represented from the same line. A different detector type shall be tested at each successive inspection interval until all structurally identical device types qualified on the same or different detail specification from the same qualified line have been tested. When none of the inspection lots passing group A of the first lot submitted contain the detector type which is due to be tested, the samples for inspection shall be chosen from these types in the inspection lots being tested which have not been used for the longest time for group C inspection. The next lot which contains the skipped type shall be subjected to group C inspection as part of its quality conformance inspection. Samples chosen for Group C shall have passed both Groups A and B.

4.5.6 Groups B and C end points. Post-tested end points specified in the detailed specification shall be measured for each detector of the sample after completion of all specified tests in the subgroup. End-point test measurements shall be performed within the time period specified after sample units have been subjected to and removed from required tests. Additional measurements may be made at the discretion of the manufacturer. At the end of each group B and C subgroup, end-point measurements shall include visual inspection without magnification to assure marking on each semiconductor tested is legible and complete. Damage to marking caused by mechanical fixturing or handling during test shall not be cause for lot rejection but devices so damaged shall be individually remarked prior to shipment. End points to be used for performance assessment shall be selected from table VIII.

TABLE VIII. End points.

End-point electrical measurements	Paragraph number	
	Requirement	Test method
Spectral responsivity and quantum efficiency	3.5.1.1	4.6.2.1
Capacitance	3.5.1.2	4.6.2.2
Reverse current leakage and breakdown voltage	3.5.1.3	4.6.2.3
Dark current	3.5.1.4	4.6.2.4
Responsivity linearity	3.5.1.5	4.6.2.5
Pulse response	3.5.1.6	4.6.2.6
Analog bandwidth	3.5.1.7	4.6.2.7
Noise equivalent power	3.5.1.8	4.6.2.8

4.5.7 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-S-19491.

4.6 Methods of inspection.

4.6.1 Physical property inspection.

4.6.1.1. Visual and mechanial inspection. Detectors shall be inspected in accordance with method 2071 of MIL-STD-750.

4.6.1.2 Physical dimensions (see 3.4.2). Detectors shall be inspected in accordance with method 2066 of MIL-STD-750 for physical conformance with the requirements of this specification. Critical dimensions to be used in this inspection shall be as specified (see 3.1).

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4.6.2 Optical/electrical. Tests which require optical inputs shall utilize optical sources as specified in DOD-S-24622.

4.6.2.1 Spectral responsivity and quantum efficiency (see 3.5.1.1). Spectral responsivity (R) measurements shall be made on the detector at the bias voltage level, ambient operating temperture level, and sinusoidal modulation frequency specified (see 3.1). A minimum of three wavelengths, representing approximately the minimum, center and maximum wavelengths of the specified detector wavelength band, as specified (see 3.1) shall be used for these measurements. Three optical input power levels at each wavelength, as specified (see 3.1), shall be used for each measurement. This measurement data is summarized in table IX (see figure 1).

TABLE IX. Measurement data.

Fixed conditions	Responsivity (amperes/watt)								
	λ_1			λ_2			λ_3		
	ϕ_1	ϕ_2	ϕ_3	ϕ_1	ϕ_2	ϕ_3	ϕ_1	ϕ_2	ϕ_3
V_{Bias} Top fm	$R_{\lambda_1\phi_1}$	$R_{\lambda_1\phi_2}$	$R_{\lambda_1\phi_3}$	$R_{\lambda_2\phi_1}$	$R_{\lambda_2\phi_2}$	$R_{\lambda_2\phi_3}$	$R_{\lambda_3\phi_1}$	$R_{\lambda_3\phi_2}$	$R_{\lambda_3\phi_3}$

Where: λ is the specified wavelength
ϕ is the specified incident optical power level
Top is the specified detector ambient operating temperature
fm is the specified optical power modulation frequency

Pigtailed LED's of the specified wavelengths shall be sinusoidally modulated at the specified frequency, their average pigtail optical output power shall be measured, and their output pigtail shall be coupled to the input pigtail of the detector. The input power to the detector window or pigtail shall be calculated as the difference between the LED pigtail output power and the interconnection loss. A connector of known loss characteristics or a splice may be used for this interconnection. The photo-induced diode output current shall be measured at all wavelengths and optical power input levels denoted in table IX. The calculated responsivity in amperes/watt (or more appropriately in microamperes/microwatt) shall be compared with and exceed the responsivity values specified (see 3.1) (at each λ and ϕ, respectively) for detector acceptance. Quantum efficiency (η) shall be calculated as $\eta = 125 \frac{R}{\lambda}$ where η is in percent, R in amperes/watt and λ is in micrometers. η shall meet the value specified (see 3.1).

4.6.2.2 Capacitance (see 3.5.1.2). Detector junction capacity shall be measured in accordance with method 4001 of MIL-STD-750. The detector bias voltage level, ambient operating temperature and electrical modulation frequency and levels shall be as specified (see 3.1). The diode shall be electrically wired in its normal circuit configuration when making this measurement (see figure 1).

4.6.2.3 Reverse current leakage and breakdown voltage (see 3.5.1.3). The reverse current leakage through the detector, at the specified breakdown voltage and ambient operating temperature, shall be measured using method 3061 of MIL-STD-750. Only the dc method for reverse current measurement shall be used (see figure 1).

4.6.2.4 Dark current (see 3.5.1.4). Detector dark current shall be measured at the bias voltage and ambient operating temperature specified (see 3.1), using method 3061 of MIL-STD-750. Bias condition C shall be used and the voltage source shown shall be adjusted until V_{EB} corresponds to the specified bias voltage. I_{EB} will then measure the detector dark current when no radiant flux enters the detector pigtail (see figure 1).

4.6.2.5 Responsivity linearity (see 3.5.1.5). The responsivity linearity shall be measured at the wavelength, ambient operating temperature, modulation frequency and bias voltage specified (see 3.1). The optical output power from the LED pigtail shall be calibrated as a function of drive level prior to coupling it to the detector window or pigtail. Interconnection loss shall be subtracted to obtain the power

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level at the input to the detector. The modulated LED shall be swept from minimum output to the maximum safe output power capability of either the source or detector, whichever is lower. Detector output current readings shall be taken at suitable input power intervals to permit the plotting of a relatively smooth relationship between output current and input power. The linearity of this plot shall not be less than the responsivity linearity specified (see 3.1) between specified radiant power levels (see figure 1).

4.6.2.6 Pulse response (see 3.5.1.6). Detector optical pulse response shall be measured at the wavelength, ambient operating temperature, bias voltage and pulse repetition frequency (PRF) specified (see 3.1). The pigtailed LED source shall be driven by an output level stabilized, square wave (50 percent duty factor), electrical pulse generator at the specified PRF. The LED and PIN window or pigtail shall be coupled through an optical attenuator. The detector output current pulse waveform shall be displayed on an oscilloscope of adequate response. The pulse rise time measured between 10% and 90% maximum pulse power amplitude points, and the pulse fall time measured between 90% and 10% maximum pulse power amplitude points shall not exceed the level specified in the applicable detail specification sheet (see 3.1)(see figure 1).

4.6.2.7 Analog bandwidth (3.5.1.7). The detector bandwidth or frequency response shall be measured as in 4.6.2.6. A calibrated sinusoidal signal generator with constant or calibrated output voltage shall be substituted for the pulse generator of 4.6.2.6. The LED and PIN pigtails or windows shall be coupled together through an attenuator and the detector output current measured at a low signal frequency (on the order of 1 kHz). The signal frequency shall be increased until the measured detector current has dropped in amplitude to 0.707 of the value obtained at the former low signal frequency. The frequency obtained at this point is the half-power or 3 decibels (dB) down frequency and represents the analog bandwidth limit. This bandwidth frequency shall not be less than the bandwidth specified (see 3.1) (see figure 1).

4.6.2.8 Noise equivalent power (see 3.5.1.8). The detector noise equivalent power shall be measured at the wavelength, ambient operating temperture, bias voltage and bandwidth specified (see 3.1). The LED shall be pulsed as in 4.6.2.6 so as to provide a square wave optical output of the modulation frequency rate specified (see 3.1) (see figure 2). A bandwidth of 1 hertz shall be used with this modulation frequency (here assummed to be 1 kHz unless otherwise specified). The source circuit is composed of a 1 kHz voltage generator and potentiometer which drives an LED optical element to provide variable intensity light output modulated at 1 kHz. This variable intensity light from the source circuit is then passed through input optics for monitoring optical power levels before being focussed upon the photodetector diode (via a pigtail optical fiber connected with the diode). The photodetector diode produces an electrical signal in the detector circuit with power proportional to the adjustable resistance value set for the potentiometer of the source circuit. The majority of this electrical signal in the detector circuit is modulated at 1 kHz, but the nonlinearities and noise characteristics of the photodetector introduce a small amount of random noise at frequencies other than 1 kHz. This superimposed voltage content of 1 kHz electrical signal and electrical noise is propagated through a low noise amplifier (A_1) operating in the broadband range of 100 Hz to 1 MHz (hence, any previously generated circuit noise which exists with frequencies outside this bandwidth is removed by the amplifier (A_1)). The output signal voltage from amplifier A_3 is then passed onto the NEP measurement circuit for determination of the signal and noise power fractions. The measurement circuit includes a switch to permit monitoring of two separate channels during different time intervals. One channel has an amplifier (A_3) of unity gain which acts as a 1,000 Hz rejection filter for measurement of the noise signal level (the noise is that power which exists at all frequencies other than 1,000 Hz). The other channel has an amplifier (A_2) of unity gain which filters out all frequencies except that of the true signal at 1,000 Hz. Flipping of the switch state enables comparison of the signal power level (V_S, I_S) against the noise power level (V_N, I_N) since the value of the resistor is the same for each leg of the detector circuit where V_S and I_S are the observed voltage and current levels, respectively, in the true signal channel; and V_N and I_N denote the observed voltage and current levels, respectively, in the noise channel.

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The amount of random noise generated in the detector circuit will become relatively large compared to the 1 kHz signal power level as the LED optical power is decreased to successively lower outputs. This is equivalent to expecting smaller signal-to-noise ratios in the output from electrical devices as the peak voltage signal decreases towards zero. It is experimentally found that the power flowing through the noise branch will equal the power flowing through the 1 kHz signal branch for some sufficiently low LED optical power output level. This low optical power output P_{IN} defines the NEP of the photodetector diode in terms of the relation:

$$NEP = 1 \times 10^{-2} \; P_{IN} \; (watts/\sqrt{Hz})$$

Hence, the average power output from the LED equals NEP in watts per square-root hertz when I_S equals I_N.

4.6.3 Environmental inspections.

4.6.3.1 High temperature (nonoperating) life (LTPD) (see 3.5.2.1). Optical detectors shall be tested to method 1032 of MIL-STD-750 to determine compliance with the specified LTPD for the nonoperating storage condition. Exposure time and temperature shall be 24 hours at +105°C (QLA devices), and 24 hours at +105°C (QLB devices) for the screening test in table IV; 340 hours at +105°C (QLA devices), and 340 hours at +105°C (QLB devices) for the group B quality conformance test in table VI.

4.6.3.2 Thermal shock (temperature cycling) (see 3.5.2.2). Detectors shall be tested to test condition A-1 of method 107 of MIL-STD-202. Except that the lower test temperature shall be -55°C and the higher temperatures shall be +105°C for QLA devices and +105°C for QLB devices. Detector mounting for the test exposure shall be specified (see 3.1).

4.6.3.3 Constant acceleration (see 3.5.2.3). Detectors shall be tested to method 2006 of MIL-STD-750. A centrifugal force of 20,000 G's shall be applied in any sequence in each of the six axial directions.

4.6.3.4 Burn-in (see 3.5.2.4). Detectors shall be tested to test condition A of method 1038 of MIL-STD-750. The detector shall be operated at 80 percent of maximum rated reverse voltage for 48 hours at a temperture of +105°C for QLA devices and +105°C for QLB devices.

4.6.3.5 Hermetic seal (see 3.5.2.5).

4.6.3.5.1 Fine leakage rate. QLA detectors shall be tested to test condition H of method 1071 of MIL-STD-750. QLB detectors shall omit this test.

4.6.3.5.2 Gross leakage rate. QLA detectors shall omit this test. QLB detectors shall be tested only to gross leak step 1 procedure of test condition C, method 1071 of MIL-STD-750. Fluid test temperature shall be +105°C.

4.6.3.6 Solderability (see 3.5.2.6). All detector external electrical leads shall be tested to method 208 of MIL-STD-202, except that the time period for aging shall be 12 hours minimum.

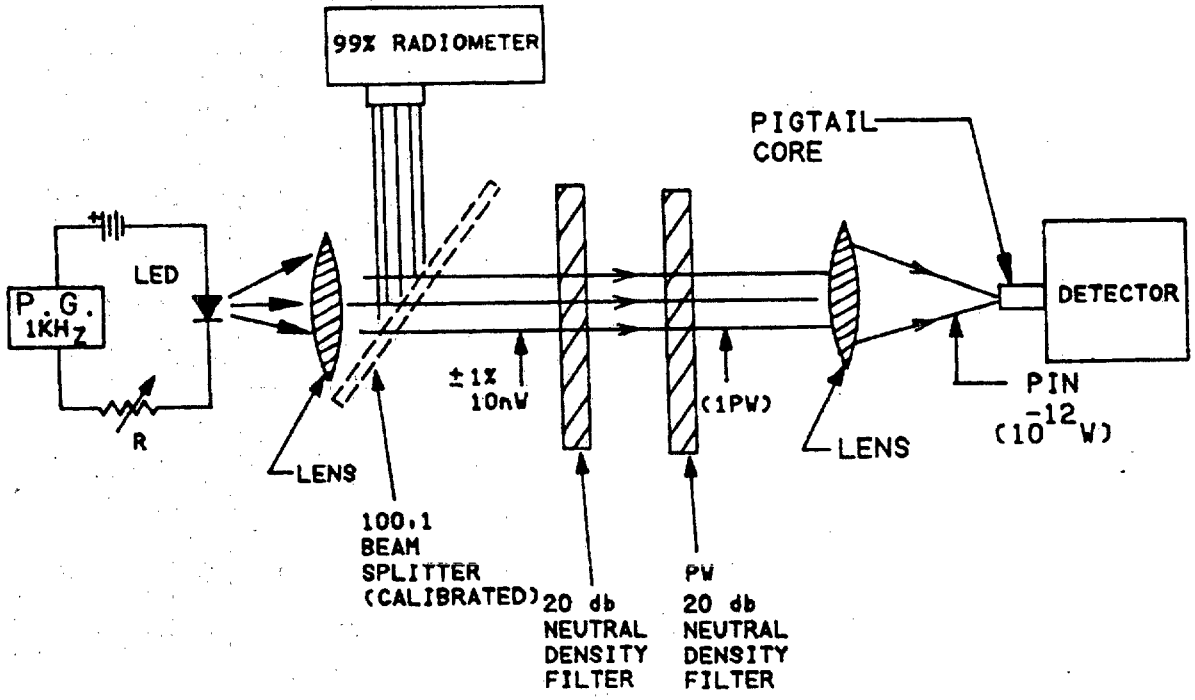
4.6.3.7 Resistance to solvents (see 3.5.2.7). Detectors shall be tested to method 1022 of MIL-STD-750.

4.6.3.8 Decap internal visual design verification (see 3.5.2.8). Detectors shall be tested to method 2075 of MIL-STD-750. Particular attention shall be paid to the pigtail/die attachment region supported by photographic documentation as described in method 2075 of MIL-STD-750.

4.6.3.9 Bond strength (see 3.5.2.9). Detectors shall be tested to test condition A, method 2037 of MIL-STD-750. Bond failure criteria shall be as stated in method 2037 of MIL-STD-750.

4.6.3.10 Thermal shock (glass strain) (see 3.5.2.10). QLA detectors shall be tested to test condition A, method 1056 of MIL-STD-750. QLB detectors shall be similarly tested but to a high temperature limit of +105°C. Distilled water shall be used as the immersion fluid.

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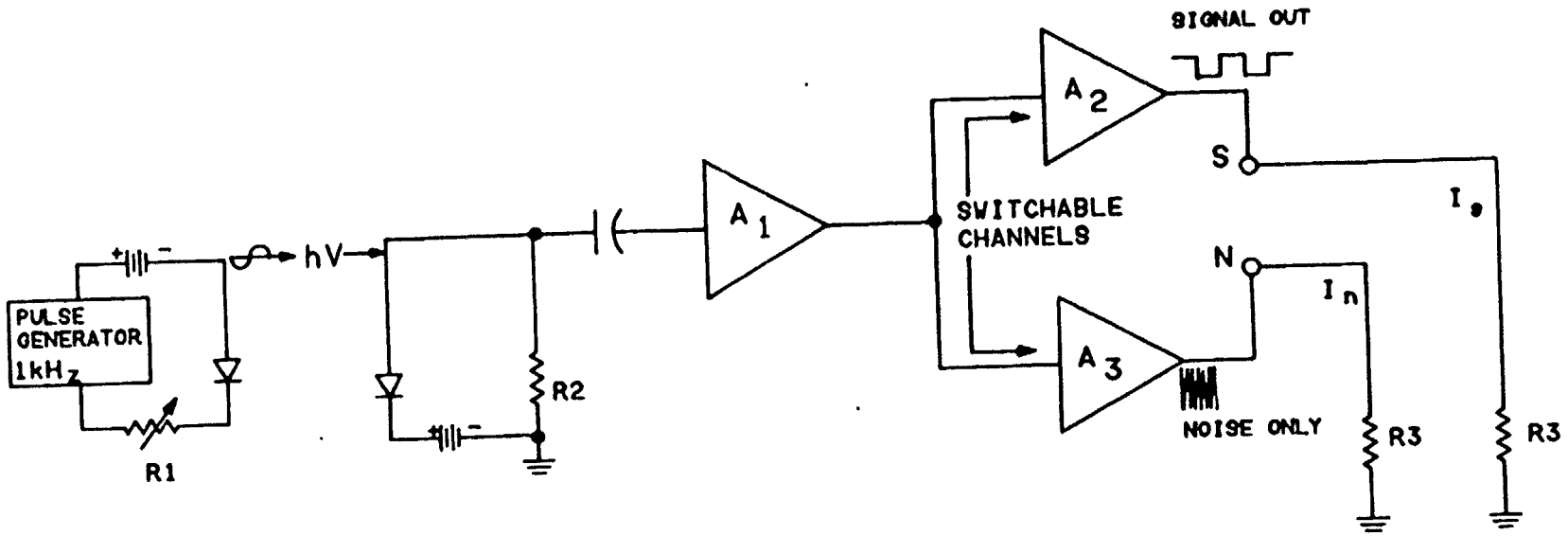
NOTE: NEP's may be as small as 10^{-14} W/$\sqrt{Hz}$, the level of power needed to the detector pigtail (P_{IN}) is of the order of 1 PW. That is,

P_{IN} = (NEP) ($\sqrt{BW_n}$), and we are using a noise bandwidth of about 10 kHz or

P_{IN} = (10^{-14}) ($\sqrt{10{,}000}$) = 10^{-12} watts.

FIGURE 1. Photodetector input optics.

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NOTES:

1. A_1, low noise, broad band linear amplifier, 100 Hz to 1 MHz.
2. A_2, 1 K_c $\pm$0.5 Hz band pass amplifier, unity gain 1 KHz amplifier.
3. A_3, 100 Hz to 10 KHz bandpass amplifier with 10 Hz notch centered at 1 KHz, unity gain.
4. P_{IN} = NEP when $V_{N(rms)} \approx V_{S(peak)}$.
5. Use RMS reading voltmeter for V_N measurement and peak reading voltmeter for V_S.
6. NEP in Watts/$\sqrt{Hz}$ = $\frac{P_{IN}}{\sqrt{BW_n}}$ = $\frac{P_{IN}}{\sqrt{10{,}000-100}}$ = $\frac{P_{IN}}{\sqrt{9900}}$ = 1×10^{-2} P_{IN} W/$\sqrt{Hz}$

FIGURE 2. NEP measurement method.

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4.6.3.11 Terminal strength (see 3.5.2.11).

4.6.3.11.1 Tension (see 3.5.2.11.1). Detectors shall be tested to test condition A, method 2036 of MIL-STD-750 using a weight of 4 ounces for 30 seconds.

4.6.3.11.2 Torque (see 3.5.2.11.2). Detectors shall be tested to test condition D1, method 2036 of MIL-STD-750 applying a torque load of 1 pound-inch for 30 seconds both clockwise and counterclockwise to the terminal.

4.6.3.12 Moisture resistance (see 3.5.2.12). Detectors shall be tested to method 1021 of MIL-STD-750. The stated initial conditioning (lead-fatigue) test of method 1021 shall be applied to the fiber pigtail and all electrical leads of the detector.

4.6.3.13 Shock (see 3.5.2.13). Detectors shall be tested to method of 2016 of MIL-STD-750. The detector shall be subjected to shocks of 1,500 G's for durations of 0.5 milliseconds each. Five shocks in each of the X1, Y1 and Z1 directions shall be given.

4.6.3.14 Vibration, variable frequency (see 3.5.2.14). Detectors shall be tested to method 2056 of MIL-STD-750.

4.6.3.15 Salt atmosphere (corrosion) (see 3.5.2.15). Detectors shall be tested to method 1041 of MIL-STD-750.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19491.

6. NOTES

6.1 Intended use. Optical detectors specified herein and in the specification sheets are of the PIN type and are intended for use in fiber optic analog and digital communication system applications.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Part number (see 1.2.1).
b. The applicable detail specification number, if applicable.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3	Certificate of compliance	DI-E-2121	---
4.1.1	Calibration system description	DI-R-7064	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 Waiver of requirement. The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement, may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the

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Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-24620) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 55Z3, DOD Standardization Program and Documents Division, Washington, DC 20362; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center, DESC-EQ, Dayton, Ohio 45444.

6.4 Definitions.

6.4.1 Dark current. Dark current is the photodiode current that flows through the biasing circuit when no light is incident on the photodiode.

6.4.2 Δ delta. Delta (Δ) is the variation in the level of an end-point parameter during the course of a test exposure. Normally, this applies to the "burn-in" screening inspection.

6.4.3 End points. End points are the parameters and limits specified which shall be met before and after the test exposure for optical detector acceptance.

6.4.4. λ lambda. λ is a symbol defined as LTPD per 1000 hours. This symbol is not to be confused with λp, the peak wavelength.

6.4.5 LTPD. LTPD (lot tolerance percent defective) is a method used for statistical sampling of parts being tested for qualification acceptance. LTPD has the form of a series where the numerical value selected represents the minimum percent of samples tested which rejects the inspection lot. Additional details are found in the appendix of this specification.

6.4.6 Noise equivalent power. Noise equivalent power (NEP) is a measure of the photodiode sensitivity. NEP represents the minimum incident optical signal power required to generate a photocurrent equal to the total photodiode noise current, i.e., a current signal to noise ratio of unity.

6.4.7 Pigtail fiber. The pigtail is a short length of optical fiber, integrally attached to the detector package at the manufacturer's plant, and alined with the detector's sensing element for maximum optical power incident thereupon.

6.4.8 Quantum efficiency. Quantum efficiency is a measure of the photoelectric conversion capability of the detector. Quantum efficiency represents the fraction of electron-hole charge pairs created per incident photon and is a figure of merit for the diode.

6.4.9 Responsivity. Detector responsivity is the wavelength-related ability of the detector to convert input optical power to output electrical current.

6.4.10 Responsivity linearity. Responsivity linearity is a measure of detector's ability to maintain a constant responsivity level over the entire usable range of optical signal input power at a given wavelength.

6.4.11 Unit container. The unit container is the package used as the initial protection for detector delivery.

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APPENDIX

STATISTICAL SAMPLING, LIFE TEST PROCEDURES

10. SCOPE

10.1 This appendix contains statistical sampling and life test procedures used with fiber optic detectors. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance only.

20. APPLICABLE DOCUMENTS

20.1 Government document.

MIL-HDBK-H53 - Guide for Attribute Lot Sampling Inspection and MIL-STD-105.

30. GENERAL

30.1 Definitions. The following definitions shall apply for all statistical sampling procedures:

a. LTPD series. The lot tolerance percent defective (LTPD) series is defined as the following decreasing series of LTPD or lambda (λ) values: 50, 30, 20, 15, 10, 7, 5, 3, 2, 1.5, 1, 0.7, 0.5, 0.3, 0.2, 0.15, 0.1.
b. Tightened inspection. Tightened inspection is defined as inspection performed using the next LTPD or lambda value in the LTPD series lower than that specified.
c. Acceptance number (c). The acceptance number is defined as an integral number associated with the selected sample size which determines the maximum number of defectives permitted for that sample size.
d. Rejection number (r). Rejection number is defined as one plus the acceptance number.
e. Lambda (λ). Lambda is defined as LTPD per 1,000 hours.

30.2 Symbols. The following symbols shall apply for all statistical sampling procedures:

a. LTPD - Lot Tolerance Percent Defective
b. λ - Lambda
c. c - Acceptance number
d. r - Rejection number
e. n - Sample size

40. STATISTICAL SAMPLING PROCEDURES AND TABLE - LTPD METHOD

40.1 General. Statistical sampling shall be conducted using the LTPD method. The LTPD method as specified herein is a double sampling plan which provides a high degree of assurance that a lot having a proportion defective equal to or greater than the specified LTPD value will not be accepted. The choice of any one procedure is optional. The procedures specified herein are suitable for all qualification or quality conformance requirements, but are not suitable if the objective of the inspection is to determine that the proportion defective in the lot represented is greater than the specified LTPD value since the assurance for that purpose is normally only 10 percent (see 6.3).

40.1.1 Selection of samples. Samples shall be randomly selected from the inspection lot or inspection sublots. (For an explanation of random sampling, see MIL-HDBK-H53, Section 13.)

40.1.1.1 Identification of samples for quality conformance tests. The authorized Government Quality Assurance representative may, at his option, mark or authorize the marking of each sample to be subjected to qualification or quality conformance inspection in order to distinguish these units from those not intended for Government conformance inspection. This additional marking shall in no way interfere with or obscure the marking required (see 3.6).

40.1.2 Failures. Failure of a unit for one or more tests of a subgroup shall be charged as a single failure.

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APPENDIX

40.2 Single-lot sampling method. Quality conformance inspection information (sample sizes and number of observed defectives) shall be accumulated from a single inspection lot to demonstrate conformance to the individual subgroup criteria.

40.2.1 Sample size. The sample size for each subgroup shall be determined from table X and shall meet the specified LTPD or lambda. The manufacturer may, at his option, select a sample size greater than that required; however, the number of failures permitted shall not exceed the acceptance number associated with the required sample size in table X.

40.2.2 Acceptance procedure. For the first sampling, an acceptance number shall be chosen and the associated number of sampling devices for the specified LTPD selected and tested (see 40.2.1). If the observed number of defectives from the first sample is less than or equal to the preselected acceptance number, the lot shall be accepted. If the observed number of defectives exceeds the preselected acceptance number, an additional sample may be chosen such that the total sample complies with 40.2.3.

40.2.3 Additional sample. The manufacturer may add an additional quantity to the initial sample, but this may be done only once for any subgroup and the added samples shall be subjected to all the tests within the subgroup. The total sample size (initial and added samples) shall be determined by a new acceptance number selected from table X.

40.2.4 Multiple criteria. Except where otherwise specified, when a subgroup contains more than one acceptance criterion, the entire sample for a subgroup shall be used for all criteria within the subgroup. In table X, the acceptance number shall be that one associated with the largest sample size in the appropriate LTPD column which is less than or equal to the sample size used.

40.2.5 One-hundred percent inspection. Inspection of 100 percent of the lot shall be allowed, at the option of the manufacturer for any or all subgroups other than those which are considered "destructive." The maximum observed percent defective for the inspection lot shall not exceed the specified LTPD or λ value. Devices that fail any test shall be removed from the lot.

40.2.6 Disposition of failed lot. A lot that fails quality conformance inspection may, at the option of the manufacturer be screened for defectives or reworked and resubmitted for reinspection (see 4.3.5).

50. LIFE TEST

50.1 General. Life tests shall be conducted in accordance with the procedures in this section. Life tests performed on devices at or within their maximum ratings shall be considered nondestructive. If a lot is made up of a collection of sublots, each sublot shall pass all applicable electrical end points as specified.

50.2 Selection of samples. Samples for life tests shall be selected at random from the inspection lot (see 40.1.1). The sample size for a 1,000-hour test shall be chosen by the manufacturer from table IX from the column under the specified λ. The acceptance number shall be the one associated with the particular sample size chosen.

50.3 Failures. A semiconductor device which exceeds one or more of the end-point limits specified for life test at any specified or other reading interval shall be considered a failure and shall not be considered acceptable at any subsequent reading interval. For the purpose of computing device hours, the test-time hours credited to a failed device shall not exceed the test time associated with the last measurement time that the device was observed to be within the specified end-point limits. If the sample fails, the test may be terminated at the discretion of the manufacturer.

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TABLE X. LTPD Sampling plans. 1/ 2/

Minimum size of sample to be tested to assure, with a 90 percent confidence, that a lot having percent-defective equal to the specified LTPD will not be accepted (single sample).

Max Percent Defective (LTPD) or x	50	30	20	15	10	7	5	3	2	1.5	1	0.7	0.5	0.3	0.2	0.15	0.1
Acceptance Number (c) (r = c + 1)	Minimum Sample Sizes (For device-hours required for life test, multiply by 1000)																
0	5 (1.03)	8 (0.64)	11 (0.46)	15 (0.34)	22 (0.23)	32 (0.16)	45 (0.11)	76 (0.07)	116 (0.04)	153 (0.03)	231 (0.02)	328 (0.02)	461 (0.01)	767 (0.007)	1152 (0.005)	1534 (0.003)	2303 (0.002)
1	8 (4.4)	13 (2.7)	18 (2.0)	25 (1.4)	38 (0.94)	55 (0.65)	77 (0.46)	129 (0.28)	195 (0.18)	258 (0.14)	390 (0.09)	555 (0.06)	778 (0.045)	1296 (0.027)	1946 (0.018)	2592 (0.013)	3891 (0.009)
2	11 (7.4)	18 (4.5)	25 (3.4)	34 (2.24)	52 (1.6)	75 (1.1)	105 (0.78)	176 (0.47)	266 (0.31)	354 (0.23)	533 (0.15)	759 (0.11)	1065 (0.080)	1773 (0.045)	2662 (0.031)	3547 (0.022)	5323 (0.015)
3	13 (10.5)	22 (6.2)	32 (4.4)	43 (3.2)	65 (2.1)	94 (1.5)	132 (1.0)	221 (0.62)	333 (0.41)	444 (0.31)	668 (0.20)	953 (0.14)	1337 (0.10)	2226 (0.062)	3341 (0.041)	4452 (0.031)	6681 (0.018)
4	16 (12.3)	27 (7.3)	38 (5.3)	52 (3.9)	78 (2.6)	113 (1.8)	158 (1.3)	265 (0.75)	398 (0.50)	531 (0.37)	798 (0.25)	1140 (0.17)	1599 (0.12)	2663 (0.074)	3997 (0.049)	5327 (0.037)	7994 (0.025)
5	19 (13.8)	31 (8.4)	45 (6.0)	60 (4.4)	91 (2.9)	131 (2.0)	184 (1.4)	308 (0.85)	462 (0.57)	617 (0.42)	927 (0.28)	1323 (0.20)	1855 (0.14)	3090 (0.085)	4638 (0.056)	6181 (0.042)	9275 (0.028)
6	21 (15.6)	35 (9.4)	51 (6.6)	68 (4.9)	104 (3.2)	149 (2.2)	209 (1.6)	349 (0.94)	528 (0.62)	700 (0.47)	1054 (0.31)	1503 (0.22)	2107 (0.155)	3509 (0.093)	5267 (0.062)	7019 (0.047)	10533 (0.031)
7	24 (16.6)	39 (10.2)	57 (7.2)	77 (5.3)	116 (3.5)	166 (2.4)	234 (1.7)	390 (1.0)	589 (0.67)	783 (0.51)	1178 (0.34)	1680 (0.24)	2355 (0.17)	3922 (0.101)	5886 (0.067)	7845 (0.051)	11771 (0.034)
8	26 (18.1)	43 (10.9)	63 (7.7)	85 (5.6)	128 (3.7)	184 (2.6)	258 (1.8)	431 (1.1)	648 (0.72)	864 (0.54)	1300 (0.36)	1854 (0.25)	2599 (0.18)	4329 (0.108)	6498 (0.072)	8660 (0.054)	12995 (0.036)
9	28 (19.4)	47 (11.5)	69 (8.1)	93 (6.0)	140 (3.9)	201 (2.7)	282 (1.9)	471 (1.2)	709 (0.77)	945 (0.58)	1421 (0.38)	2027 (0.27)	2842 (0.19)	4733 (0.114)	7103 (0.077)	9468 (0.057)	14206 (0.038)
10	31 (19.9)	51 (12.1)	75 (8.4)	100 (6.3)	152 (4.1)	218 (2.9)	306 (2.0)	511 (1.2)	770 (0.80)	1025 (0.60)	1541 (0.40)	2199 (0.28)	3082 (0.20)	5133 (0.120)	7704 (0.080)	10268 (0.060)	15407 (0.040)
11	33 (21.0)	54 (12.8)	83 (8.3)	111 (6.2)	166 (4.2)	238 (2.9)	332 (2.1)	555 (1.2)	832 (0.83)	1109 (0.62)	1664 (0.42)	2378 (0.29)	3323 (0.21)	5546 (0.12)	8319 (0.083)	11092 (0.062)	16638 (0.042)
12	36 (21.4)	59 (13.0)	89 (8.6)	119 (6.5)	178 (4.3)	254 (3.0)	356 (2.2)	594 (1.3)	890 (0.86)	1187 (0.65)	1781 (0.43)	2544 (0.3)	3562 (0.22)	5936 (0.13)	8904 (0.086)	11872 (0.065)	17808 (0.043)
13	38 (22.3)	63 (13.4)	95 (8.9)	126 (6.7)	190 (4.5)	271 (3.1)	379 (2.26)	632 (1.3)	948 (0.89)	1264 (0.67)	1896 (0.44)	2709 (0.31)	3793 (0.22)	6321 (0.134)	9482 (0.089)	12643 (0.067)	18964 (0.045)
14	40 (23.1)	67 (13.8)	101 (9.2)	134 (6.9)	201 (4.6)	288 (3.2)	403 (2.3)	672 (1.4)	1007 (0.92)	1343 (0.69)	2015 (0.46)	2878 (0.32)	4029 (0.23)	6716 (0.138)	10073 (0.092)	13431 (0.069)	20146 (0.046)
15	43 (23.3)	71 (14.1)	107 (9.4)	142 (7.1)	213 (4.7)	305 (3.3)	426 (2.36)	711 (1.41)	1066 (0.94)	1422 (0.71)	2133 (0.47)	3046 (0.33)	4265 (0.235)	7108 (0.141)	10662 (0.094)	14216 (0.070)	21324 (0.047)
16	45 (24.1)	74 (14.6)	112 (9.7)	150 (7.2)	225 (4.8)	321 (3.37)	450 (2.41)	750 (1.44)	1124 (0.96)	1499 (0.72)	2249 (0.48)	3212 (0.337)	4497 (0.241)	7496 (0.144)	11244 (0.096)	14992 (0.072)	22487 (0.048)
17	47 (24.7)	79 (14.7)	118 (9.86)	158 (7.36)	236 (4.93)	338 (3.44)	473 (2.46)	788 (1.48)	1182 (0.98)	1576 (0.74)	2364 (0.49)	3377 (0.344)	4728 (0.246)	7880 (0.148)	11819 (0.098)	15759 (0.074)	23639 (0.049)
18	50 (24.9)	83 (15.0)	124 (10.0)	165 (7.54)	248 (5.02)	354 (3.51)	496 (2.51)	826 (1.51)	1239 (1.0)	1652 (0.75)	2478 (0.50)	3540 (0.351)	4956 (0.251)	8260 (0.151)	12390 (0.100)	16520 (0.075)	24780 (0.050)
19	52 (25.5)	86 (15.4)	130 (10.2)	173 (7.76)	259 (5.12)	370 (3.58)	518 (2.56)	864 (1.53)	1296 (1.02)	1728 (0.77)	2591 (0.52)	3702 (0.358)	5183 (0.256)	8638 (0.153)	12957 (0.102)	17276 (0.077)	25914 (0.051)
20	54 (26.1)	90 (15.6)	135 (10.4)	180 (7.82)	271 (5.19)	386 (3.65)	541 (2.60)	902 (1.56)	1353 (1.04)	1803 (0.78)	2705 (0.52)	3864 (0.364)	5410 (0.260)	9017 (0.156)	13526 (0.104)	18034 (0.078)	27051 (0.052)
25	65 (27.0)	109 (16.1)	163 (10.8)	217 (8.08)	326 (5.38)	466 (3.76)	652 (2.69)	1086 (1.61)	1629 (1.08)	2173 (0.807)	3259 (0.538)	4656 (0.376)	6518 (0.269)	10863 (0.161)	16295 (0.108)	21726 (0.081)	32589 (0.054)

1/ Sample sizes are based upon the Poisson exponential binomial limit.
2/ The minimum quality (approximate AQL) required to accept (on the average) 19 of 20 lots is shown in parenthesis for information only.

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50.4 Life-test time and sample size. Whenever a lambda (λ) is specified, the life-test time shall be 1,000 hours initially. Once a lot has passed the 1,000-hour test, life tests with minimums of 340 hours and maximums of 2,000 hours may be initiated for new lots provided that 180 days have not elapsed since a 1,000-hour life test on the same device type on the same structurally identical group (see 4.3.1.1.2). If 180 days have elapsed, the new lot shall pass a 1,000-hour life test. The sample size for a life-test time other than 1,000 hours shall be chosen according to the relationship of inverse proportionality between test time and sample size, such that the total unit test hours accumulated (sample sizes multiplied by test hours) equal the amount that would have been chosen for the 1,000-hour life test, had it been performed. The acceptance number shall also be determined from the sample size associated with the same 1,000-hour test, had it been performed. The lot shall be accepted if the number of failures at the end of the test period does not exceed the acceptance number.

50.5 Procedure to be used if number of observed failures exceeds the acceptance number. In the event that the number of failures observed on life test exceeds the acceptance number, the manufacturer shall choose one of the following options: (1) discontinue the life test, screen or rework and resubmit per paragraph 40.2.6, (2) add additional samples in accordance with 50.5.1, or (3) extend the test time to 1,000 hours in accordance with 50.5.2, if a test time less than 1,000 hours was originally chosen. Only one of these options shall be used for a given submission, and this option shall be used only once.

50.5.1 Additional samples. When this option is chosen, a new total sample size (initial plus added) shall be chosen by the manufacturer from table X from the column under the specified λ. A quantity of additional units sufficient to increase the sample to the newly chosen total sample size shall be selected from the lot. A new acceptance number shall be determined and shall be the one associated with the new total sample size chosen. The added sample shall be subjected to the same life-test conditions and time period as the initial sample. If the total observed number of defectives (initial plus added) does not exceed the acceptance number for the total sample, the lot shall be accepted. If the observed number of defectives exceeds this acceptance number, the lot shall not be accepted but may be resubmitted (see 40.2.6).

50.5.2 Extension of life-test period. If a life test time period less than 1,000 hours is being used and the number of failures observed in the inital sample exceeds the acceptance number, the manufacturer may, in lieu of adding additional samples, choose to extend the test time of the entire initial sample to 1,000 hours and determine a new acceptance number from table X. The new acceptance number shall be that one associated with the largest sample size in the specified column which is less than or equal to the sample size on test. A device which is a failure at the initial reading interval shall not be considered acceptable at the 1,000 hour reading interval. If the observed number of defectives at 1,000 hours does not exceed the new acceptance number, the lot shall be accepted. If the observed number of defectives exceeds this acceptance number, the lot shall not be accepted, but may be resubmitted (see 40.2.6).

50.5.3 Failure of life test. If a lot fails to meet life-test requirements (including resubmission per paragraph 40.2.6, if elected) such that it is eliminated or withdrawn from further quality conformance inspection consideration, then 1,000-hour life tests shall be required until three successive lots have passed the specified life tests. Then life testing in accordance with 50.2 and 50.4 may be resumed.

50.6 Failure of life test. If a lot fails to meet life-test requirements (including submission per paragraph 40.2.6, if elected) such that it is eliminated or withdrawn from further quality conformance inspection consideration, then a 1,000-hour life test shall be required for the failed subgroup until three successive lots have passed. If group B or C (table IV or V) does not require 1,000-hour testing, then the specified life test, if other than 1,000 hours, shall be required for three successive lots.

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